

A STUDY OF RANDOM MATRICES

AN HONORS THESIS

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Abstract

[TODO] *Write the abstract once the body is finalized. Suggested skeleton: (1) Open with a one-sentence framing of random matrix theory and why sample covariance matrices matter in high-dimensional statistics. (2) State the scope: we focus on Gaussian sample covariance matrices in Johnstone’s spiked model. (3) Outline the journey: foundational probability and linear algebra, sample covariance matrices and the Wishart distribution, low-dimensional fluctuation behavior, the high-dimensional Marchenko-Pastur and Tracy-Widom laws, the spiked covariance model and the BBP phase transition, eigenvector alignment beyond the threshold, and a sketch of the connection to spectral clustering. (4) Note the role of Monte Carlo simulations in illustrating each phase of the argument. (5) Close with the takeaway: the contrast between strong and weak spikes produces a clean phase transition in both eigenvalue location and eigenvector informativeness, with implications for spectral clustering. Avoid em dashes; use clean academic prose.*

Acknowledgements

[TODO] *Acknowledgements paragraph. Suggested mentions: Professor Gustavo Didier (advisor) for the weekly meetings, the reading recommendations (Yao, Zheng, and Bai 2015; Féral and Piché 2009), and the patient guidance through the spiked covariance literature. The second reader (TBD) for serving on the committee. Optional: family, friends, and the Tulane Department of Mathematics.*

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1 Introduction

[TODO] *Open with a one-paragraph hook: random matrix theory began in the 1950s in nuclear physics and has since become central to high-dimensional statistics and data science. The fundamental object in multivariate statistics is the sample covariance matrix, whose eigenvalues describe how variance is distributed across principal directions. When the number of variables p is small compared to the number of samples n , the sample covariance is well behaved. When p and n are comparable, classical intuitions break down and qualitatively new phenomena appear.*

[TODO] *Second paragraph: scope. This thesis examines the spectral behavior of Gaussian sample covariance matrices in two regimes: (i) the low-dimensional regime where p is fixed and $n \rightarrow \infty$; (ii) the high-dimensional regime where $p/n \rightarrow y \in (0, \infty)$. In the high-dimensional regime, we focus on Johnstone’s spiked model*

$$\Sigma = \text{diag}(\alpha_1, \dots, \alpha_m, 1, \dots, 1),$$

in which the population covariance has finitely many distinguished eigenvalues (“spikes”) above a flat baseline of ones.

[TODO] *Third paragraph: main story. The central phenomenon is the BBP phase transition: a spike $\alpha > 1 + \sqrt{y}$ creates an isolated outlier in the sample spectrum with informative limiting eigenvector, while a sub-threshold spike is absorbed into the bulk and its eigenvector becomes uninformative. This contrast underpins when methods like principal components analysis and spectral clustering succeed in high dimensions.*

[TODO] *Fourth paragraph: roadmap. Section 2 covers the probability foundations needed for the rest of the thesis. Section 3 develops sample covariance matrices, the Wishart distribution, the low-dimensional Gaussian and GOE fluctuation regimes, and the high-dimensional Marchenko-Pastur and Tracy-Widom laws. Section 4 introduces the spiked covariance model and proves the BBP phase transition for eigenvalues, discusses fluctuations of spiked eigenvalues, derives the eigenvector alignment formula, and sketches the connection to spectral clustering. Section 5 concludes with a summary and open questions. Three appendices collect auxiliary results, simulation methodology, and selected code.*

2 Probability Foundations

[TODO] Opening paragraph: motivation. The high-dimensional phenomena we will study (Marchenko-Pastur, Tracy-Widom, BBP transition) build on a few foundational ideas from classical probability: random variables and distributions, modes of convergence, the law of large numbers and central limit theorem, and random vectors with multivariate normal distributions. This section reviews the minimum we need; full development is in [Casella and Berger \(2002\)](#); [Billingsley \(1995\)](#).

2.1 Random Variables and Distributions

[TODO] Content to fill in (from v5 + Joseph Section 2.1): - Definition: random variable $X : \Omega \rightarrow \mathbb{R}$. - Definition: CDF $F_X(x) = \mathbb{P}(X \leq x)$ and its three characterizing properties (limits, right-continuity, monotonicity). - Definition: discrete vs. absolutely continuous random variables. - Examples: Bernoulli, Poisson, Normal, Gamma (densities). Cite ([Casella and Berger, 2002](#), §1.2 and §5.5).

2.2 Modes of Convergence and the Law of Large Numbers

[TODO] Content to fill in (from v5 + Joseph Section 2.2): - Definitions: convergence almost surely, in probability, in distribution. - Implication chain: a.s. $\Rightarrow$ probability $\Rightarrow$ distribution; counterexamples to the converses ($1/n$ indicator). - Markov's inequality, Chebyshev's inequality (proofs). - Chebyshev's weak law of large numbers (proof via Chebyshev). - Kolmogorov's strong LLN (statement, cite Billingsley). - Include Figure: WLLN simulation (normal and exponential samples, sample mean vs. n). This figure exists in Joseph's thesis; we will reproduce in matplotlib with our color palette (notebook 01).

2.3 Moments and the Moment Generating Function

[TODO] Content to fill in (from v5 MGF section, condensed): - Definition: $M_X(t) = \mathbb{E}[e^{tX}]$, existence on $(-h, h)$. - Proposition: $M_X^{(n)}(0) = \mathbb{E}[X^n]$ (one-line proof, differentiate under the integral). - One worked example: $\mathcal{N}(\mu, \sigma^2)$ MGF $= e^{\mu t + \sigma^2 t^2/2}$. - Note that heavy-tailed laws (e.g., Cauchy) lack an MGF; defer characteristic functions to [Appendix A](#).

2.4 Random Vectors, the Multivariate Normal, and the CLT

[TODO] Content to fill in (from v5 + Joseph Section 2.4): - Definition: random vector $\mathbf{X} = (X_1, \dots, X_p)^\top$, mean $\boldsymbol{\mu}$, covariance $\Sigma = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top]$. - Property: Σ is positive semidefinite (proof via $v^\top \Sigma v = \text{Var}(v^\top \mathbf{X})$). - Definition: multivariate normal density. - Statement of the multivariate central limit theorem (one-line proof sketch via characteristic functions; full proof in [Appendix A](#)).

3 Sample Covariance Matrices

[TODO] *Opening paragraph: the sample covariance matrix $\hat{\Sigma}_n$ is the central object of multivariate statistics. In low dimensions (p fixed, $n \rightarrow \infty$) it converges almost surely to the population covariance Σ and its eigenvalues converge by continuity. In high dimensions ($p, n \rightarrow \infty$ with $p/n \rightarrow y > 0$) this convergence fails dramatically: the empirical spectrum spreads out into a non-trivial bulk density. This section develops both regimes, building from the foundations laid in Section 2.*

3.1 Population and Sample Covariance, Spectral Theorem, Positive Semidefiniteness

[TODO] *Content to fill in (from v5 + Joseph Section 3 intro): - Definition: sample covariance matrix $\hat{\Sigma}_n = \frac{1}{n} \sum_{i=1}^n \mathbf{X}_i \mathbf{X}_i^\top$ (mean-zero case) and the unbiased $\frac{1}{n-1}$ variant. - Matrix form: $\hat{\Sigma}_n = \frac{1}{n} \mathbf{X} \mathbf{X}^\top$ for $\mathbf{X} \in \mathbb{R}^{p \times n}$. - Spectral theorem for symmetric matrices: $A = Q \Lambda Q^\top$. - PSD characterization: A is PSD iff all eigenvalues are nonneg (proof). - Use the diagonalization to compute quadratic forms; foreshadow why eigenvalues of $\hat{\Sigma}_n$ are the meaningful quantities.*

3.2 Low Dimensions: Wishart, Eigenvalue Repulsion, and Fluctuations

[TODO] *Content to fill in (from v5 + Joseph Section 3.1):*

(a) *Almost sure convergence of the sample covariance. - Theorem: if $\mathbf{X}_i$ iid with $\mathbb{E} \mathbf{X}_1 = 0$ and $\text{Cov}(\mathbf{X}_1) = \Sigma$, then $\hat{\Sigma}_n \xrightarrow{\text{a.s.}} \Sigma$ entrywise by SLLN. - Corollary: $\lambda_i(\hat{\Sigma}_n) \xrightarrow{\text{a.s.}} \lambda_i(\Sigma)$ by continuity of eigenvalues (proof sketch via continuity of the characteristic polynomial and continuity of its roots). Cite Genzer (2025, Prop. 3.1, 3.3).*

(b) *Wishart distribution. - Definition: $\mathbf{S} = \sum_{i=1}^n \mathbf{X}_i \mathbf{X}_i^\top \sim W_p(n, \Sigma)$ for iid Gaussian $\mathbf{X}_i \sim \mathcal{N}(0, \Sigma)$. - Wishart density formula. - Theorem: for $p = 1$, $W_1(n, 1)$ collapses to χ_n^2 (proof by plugging in).*

(c) *2D Wishart and eigenvalue repulsion. - Joint density of ordered eigenvalues of $W_2(n, I_2)$ contains the factor $\lambda_2 - \lambda_1$, producing eigenvalue repulsion. - Simulation: scatter of (λ_1, λ_2) and histogram of $\lambda_2 - \lambda_1$ (notebook 02). Include figure: 2D Wishart eigenvalue scatter.*

(d) *Joint CLT for entries of $\hat{\Sigma}_n$ when $\Sigma = I_2$. - Statement and proof via multivariate CLT (variances 2, 2, 1 for diagonal and off-diagonal entries). - Consequence: matrix-valued CLT $\sqrt{n}(\hat{\Sigma}_n - I_2) \xrightarrow{d} G$ where G has GOE structure.*

(e) *Top eigenvalue fluctuation: repeated vs. distinct cases. - Repeated ($\Sigma = I_2$): $\sqrt{n}(\lambda_2(\hat{\Sigma}_n) - 1) \xrightarrow{d} \lambda_2(G)$, not Gaussian. - Distinct ($\Sigma = \text{diag}(1, 2)$): delta-method gives Gaussian fluctuations with variances 2 and 8. - Simulation: Q-Q plots for both cases (notebook 03). Include figures: Q-Q repeated, Q-Q distinct.*

(f) *Eigenvector convergence.* - Distinct case: $\widehat{\mathbf{u}}_\ell \xrightarrow{\text{a.s.}} \mathbf{e}_\ell$ with sign convention. - Repeated case: eigenvectors fail to converge; explicit counterexample with rotating O_n . This previews why spectral separation is essential for eigenvector recovery, a theme that returns in Section 4.

3.3 High Dimensions: Marchenko-Pastur and the Tracy-Widom Edge

[TODO] Content to fill in (from v5 high-dim section + Joseph Section 3.2):

(a) *The high-dimensional regime.* - Setup: $p, n \rightarrow \infty$ with $p/n \rightarrow y \in (0, \infty)$. - Empirical spectral distribution $F_{\widehat{\Sigma}_n}(\lambda) = \frac{1}{p} \#\{i : \lambda_i(\widehat{\Sigma}_n) \leq \lambda\}$.

(b) *Marchenko-Pastur law.* - Theorem (Marchenko-Pastur 1967): for null $\Sigma = I_p$ and $0 < y < 1$, $F_{\widehat{\Sigma}_n} \xrightarrow{d} \text{MP}_y$ with support $[(1 - \sqrt{y})^2, (1 + \sqrt{y})^2]$ and density $f_{\text{MP}}(\lambda) = \frac{1}{2\pi y \lambda} \sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}$. - For $y \geq 1$, an atom at zero of mass $1 - 1/y$. - Convergence of $\lambda_{\max}(\widehat{\Sigma}_n) \xrightarrow{\text{a.s.}} (1 + \sqrt{y})^2$. - Simulation: histogram against MP density at $y = 1/2$ (notebook 04). Include figure: MP density vs. simulation.

(c) *Tracy-Widom edge heuristic.* - Goal: explain why edge fluctuations live on scale $p^{-2/3}$, not $n^{-1/2}$. - Heuristic: square-root density near the edge, $f_{\text{MP}}(\lambda) \sim C(\lambda_+ - \lambda)^{1/2}$ as $\lambda \uparrow \lambda_+$, combined with the requirement that the mass to the right of $\lambda_{\max}$ be of order $1/p$, yields $\lambda_+ - \lambda_{\max} \sim p^{-2/3}$. - Statement (informal): $p^{2/3}(\lambda_{\max} - \lambda_+) \xrightarrow{d} \text{TW}_1$ after model-dependent constants. - Remark on large-deviation heuristic confirming the same scaling from left and right tails. Cite Tracy and Widom (1996); Johnstone (2001).

4 Spiked Covariance Matrices and the BBP Transition

[TODO] Opening paragraph: the Marchenko-Pastur law of Section 3.3 described the null case $\Sigma = I_p$, where all population eigenvalues are equal. In applications the population covariance carries structure: a small number of distinguished directions of high variance riding above a flat noise baseline. The spiked covariance model of [Johnstone \(2001\)](#) formalizes this,

$$\Sigma = \text{diag}(\alpha_1, \dots, \alpha_m, 1, \dots, 1),$$

with m fixed as $p \rightarrow \infty$. The central question is when these population spikes are detectable from the sample spectrum, and when the sample eigenvectors carry meaningful information about the population spike directions. The answer is a clean phase transition.

4.1 Johnstone's Spiked Model and the Spike-Location Map

[TODO] Content to fill in (from v5 + Yao-Zheng-Bai Ch. 11 + Gustavo notes): - Definition: Johnstone's spiked model $\Sigma = \text{diag}(\alpha_1, \dots, \alpha_m, 1, \dots, 1)$ with $H = \delta_1$ for the bulk limiting distribution. - Definition: spike-location map $\Psi(\alpha) = \alpha + \frac{y\alpha}{\alpha-1}$. - Compute $\Psi'(\alpha) = 1 - \frac{y}{(\alpha-1)^2}$. - Lemma: $\Psi'(\alpha) > 0 \iff |\alpha - 1| > \sqrt{y}$; for large spikes $\alpha > 1$ this becomes $\alpha > 1 + \sqrt{y}$. - Verify boundary behavior: $\Psi(1 \pm \sqrt{y}) = (1 \pm \sqrt{y})^2$, so the spike map agrees with the MP edges at the threshold.

4.2 The BBP Phase Transition for Eigenvalues

[TODO] Content to fill in: - Theorem (Yao-Zheng-Bai Corollary 11.4 specialized to Johnstone): For a large spike $\alpha_j > 1 + \sqrt{y}$,

$$\lambda_i(\widehat{\Sigma}_n) \xrightarrow{\text{a.s.}} \Psi(\alpha_j), \quad i \in J_j.$$

For $1 < \alpha_j \leq 1 + \sqrt{y}$,

$$\lambda_i(\widehat{\Sigma}_n) \xrightarrow{\text{a.s.}} (1 + \sqrt{y})^2.$$

Brief sketch (cite [Baik and Silverstein \(2006\)](#) for the proof). - Worked example: $y = 1/2$, so $1 + \sqrt{y} \approx 1.707$ and edge $(1 + \sqrt{y})^2 \approx 2.914$. For $\alpha = 2.5$, $\Psi(2.5) \approx 3.333$. For $\alpha = 1.4$, $\alpha < 1.707$ so the spike is absorbed into the edge. - Simulations (existing figures from v5): Figure: smoothed $m=1$ simulation, strong vs. weak spike ([smoothed_m1_strong_vs_weak.png](#)). Figure: raw top eigenvalue distributions ([m1_strong_spike.png](#), [m1_weak_spike.png](#)). Figure (optional): $m = 2$ cases ([m2_two_strong_spikes.png](#), [m2_mixed_spikes.png](#)). Cite [Baik et al. \(2005\)](#); [Yao et al. \(2015\)](#).

4.3 Fluctuations of Spiked Eigenvalues

[TODO] Content to fill in (from Gustavo's promised sketch + v5): - Statement (fundamental spike, qualitative): for $\alpha > 1 + \sqrt{y}$ in the Gaussian Johnstone model,

$$\sqrt{n}(\lambda_{\max}(\hat{\Sigma}_n) - \Psi(\alpha)) \xrightarrow{d} \mathcal{N}(0, \sigma_{\alpha,y}^2).$$

Cite Yao et al. (2015, Thm. 11.11) (fundamental spikes) and Féral and Péché (2009, Thm. 1.6) (cleanest diagonal case). - Heuristic: a separated eigenvalue is a smooth functional of $\hat{\Sigma}_n$, so a delta-method argument gives $\sqrt{n}$ Gaussian fluctuations (compare with Section 3.2(e)). - Non-fundamental spike: top eigenvalue attaches to the edge, $\lambda_{\max} - \lambda_+ = O_p(p^{-2/3})$, and the fluctuation is expected to be Tracy-Widom type (cite Johnstone (2001)). - Honest note: the critical case $\alpha = 1 + \sqrt{y}$ is not resolved in our current treatment; cite open literature. - Simulation: histogram and Gaussian Q-Q for strong-spike fluctuations (notebook 06).

4.4 Eigenvector Alignment Beyond the Threshold

[TODO] Content to fill in (from v5 + Yao-Zheng-Bai Thm 11.5/Cor 11.6): - Set-up: in the one-spike model, the population spike direction is $\mathbf{e}_1$. Let $\hat{\mathbf{u}}_1$ be the top sample eigenvector. The squared overlap $|\langle \hat{\mathbf{u}}_1, \mathbf{e}_1 \rangle|^2 = \cos^2 \theta$ removes the sign ambiguity. - Qualitative theorem (Yao-Zheng-Bai Thm. 11.5, Cor. 11.6): strong spike $\Rightarrow$ overlap converges to a positive constant; weak spike $\Rightarrow$ overlap converges to zero. - The one-spike Johnstone overlap formula:

$$\rho(\alpha, y) = \frac{1 - \frac{y}{(\alpha - 1)^2}}{1 + \frac{y}{\alpha - 1}}.$$

Positive precisely when $\alpha > 1 + \sqrt{y}$. - Simulation (figure already in v5/figures/): **eigenvector_overlap_dist** shows strong-spike overlaps clustered around 0.55–0.65 and weak-spike overlaps near 0. Include and discuss.

4.5 Sketch of the Connection to Spectral Clustering

[TODO] Content to fill in (from Gustavo's email + v5 + von Luxburg 2007): - Setup: data points $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^p$, kernel similarity $K_{ij} = f(\|\mathbf{x}_i - \mathbf{x}_j\|^2/p)$. - Definition: degree matrix $D = \text{diag}(K\mathbf{1})$ and normalized Laplacian-style kernel $L = nD^{-1/2}KD^{-1/2}$. - Heuristic decomposition: $L = \text{noise bulk} + \text{low-rank class signal}$, in direct analogy with $\hat{\Sigma}_n = \text{noise} + \text{spike}$. - Conclusion: class structure is recoverable from L precisely when the signal-to-noise ratio crosses an analogue of the BBP threshold; the informative eigenvectors of L then provide the clustering. Cite von Luxburg (2007) for the algorithmic side and Couillet and Benaych-Georges (2016) for the high-dimensional analysis. - Toy simulation: notebook 08, a two-class Gaussian mixture and the top non-trivial eigenvector of L .

5 Conclusion and Outlook

[TODO] Content to fill in: (a) Summary table: the strong vs. weak spike contrast, three rows: eigenvalue limit, eigenvector overlap, fluctuation scale.

— Spike strength — Eigenvalue limit — Overlap with $\mathbf{e}_1$ — Fluctuation — ————
 — Strong $\alpha > 1 + \sqrt{y}$ — $\Psi(\alpha)$
 — $\rho(\alpha, y) > 0$ — Gaussian, $\sqrt{n}$ — — Weak $1 < \alpha \leq 1 + \sqrt{y}$ — $(1 + \sqrt{y})^2$ — 0 — Edge / Tracy-Widom, $p^{-2/3}$ —

(b) One-paragraph synthesis: the BBP phase transition is the cleanest quantitative example of when high-dimensional inference recovers structure and when it does not. It explains, in a controlled toy model, why methods like PCA and spectral clustering succeed only when signal exceeds a threshold determined by the aspect ratio.

(c) Outlook: open questions to flag. - The critical case $\alpha = 1 + \sqrt{y}$. The current literature does not yet contain a clean limit theorem; cite open work. - Exact variance $\sigma_{\alpha, y}^2$ in the strong-spike CLT. - Beyond Johnstone's case: general spiked models with non-flat bulk (cite Baik and Silverstein (2006); Benaych-Georges and Nadakuditi (2011)). - Rigorous spectral clustering derivations beyond the toy model; cite Couillet and Benaych-Georges (2016).

(d) Closing sentence: the goal of this thesis was to develop the foundational picture from one end of the dimension spectrum to the other; the open questions above point to where the field is going.

A Auxiliary Results

[TODO] Content to fill in:

A.1 Borel-Cantelli Lemmas. - First Borel-Cantelli: if $\sum_n \mathbb{P}(A_n) < \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 0$. - Second Borel-Cantelli (independence version): if events are independent and $\sum_n \mathbb{P}(A_n) = \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 1$. - Used in Section 2.2 for the convergence-in-probability but not-almost-surely counterexample.

A.2 Slutsky's theorem. - Statement and a one-paragraph proof or reference to Casella and Berger (2002, Thm. 5.5.17).

A.3 Continuous mapping theorem. - Statement: if $X_n \xrightarrow{d} X$ and g is continuous on a set of $\mathbb{P}_X$ -probability one, then $g(X_n) \xrightarrow{d} g(X)$.

A.4 Multivariate delta method. - Statement and reference; used in Sections 3.2 and 4.3.

A.5 Characteristic functions and the Lévy continuity theorem. - Definition: $\varphi_X(t) = \mathbb{E}[e^{itX}]$; always exists. - Examples: $\mathcal{N}(0, 1)$ gives $e^{-t^2/2}$, Cauchy gives $e^{-|t|}$. - Lévy: $X_n \xrightarrow{d} X \iff \varphi_{X_n}(t) \rightarrow \varphi_X(t)$ pointwise, with continuity at 0 sufficient for the converse. - Proof of the classical CLT via $(1 + t/n)^n \rightarrow e^t$ expansion.

B Simulation Methodology

[TODO] Content to fill in:

B.1 General reproducibility conventions. - All notebooks use Python 3, with `numpy`, `matplotlib`, and `scipy.stats`. - Random number generation: `rng = np.random.default_rng(seed)`, with explicit `seed` at the top of every notebook. - Eigendecomposition: `np.linalg.eigh` on symmetric inputs; the top eigenpair is taken as `evals[-1]`, `evecs[:, -1]`. - Plot styling: shared helper module `notebooks/thesis_style.py` applies the six-color palette and sans-serif math labels.

B.2 Per-notebook parameter table. - Notebook 01 (LLN/CLT demos): $n \in \{100, 1000, 10000\}$, 500 trials. - Notebook 02 (Wishart and repulsion): $n = 100$, $p = 2$, 5000 trials. - Notebook 03 (fixed-dim fluctuations): $n = 500$, 5000 trials, both $\Sigma = I_2$ and $\Sigma = \text{diag}(1, 2)$. - Notebook 04 (Marchenko-Pastur & TW): $n = 1000$, $y = 1/2$, 1 trial for bulk density, 1000 trials for top eigenvalue. - Notebook 05 (spiked eigenvalues, BBP): $y = 1/2$, $n \in \{100, 200, 400, 800, 1200\}$, 200 trials per n , $\alpha \in \{1.4, 2.5\}$. - Notebook 06 (fluctuation Q-Q): $n = 1200$, $y = 1/2$, 500 trials, $\alpha = 2.5$. - Notebook 07 (eigenvector overlap): $n = 1200$, $y = 1/2$, 500 trials, $\alpha \in \{1.4, 2.5\}$. - Notebook 08 (spectral clustering toy): two-class Gaussian mixture, $n = 600$ per class, $p \in \{50, 200\}$.

B.3 Numerical caveats. - For the strong-spike fluctuation Q-Q plot, we standardize empirically (subtract sample mean, divide by sample standard deviation) because the exact variance $\sigma_{\alpha, y}^2$ requires constants from Yao et al. (2015, Thm. 11.11). - For the weak spike, we do not attempt an exact Tracy-Widom fit because the finite-sample centering and scaling constants are delicate; we report the qualitative shape of the histogram and defer the rigorous fit to future work. - The realized aspect ratio p/n differs from the target y when yn is not integer; figures use the realized ratio in the bulk-edge formula $(1 + \sqrt{p/n})^2$ for consistency.

C Selected Code Listings

[TODO] *Include 4 to 6 of the most informative snippets from the notebooks (not full notebooks). Each listing should be self-contained at the function level, captioned, and referenced from the main text where the corresponding figure appears.*

Suggested listings to include here:

- *C.1 One-spike Johnstone simulation (driving function for Section 4.2). Source: `notebooks/05_spiked`*
- *C.2 Eigenvector overlap simulation (Section 4.4). Source: `notebooks/07_eigenvector_overlap.ipynb`*
- *C.3 Strong-spike fluctuation simulation (Section 4.3). Source: `notebooks/06_fluctuation_qq_plots`*
- *C.4 Marchenko-Pastur bulk simulation (Section 3.3). Source: `notebooks/04_marchenko_pastur_trace`*

Each listing will be wrapped in a `lstlisting` environment using the `thesisPython` style defined in `style/thesis.sty`, with a caption and a `label` for cross-referencing from the main text.

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